SHORT-PAPER

MERA: Mixture of Experts with Retrieval-Augmented Representation for Modeling Diversified Stock Patterns

YUJUN LIU, Tsinghua University, Beijing, China

CHENHUI SONG

PEIYUAN LIU, Tsinghua University, Beijing, China

NAIQI LI, Tsinghua University, Beijing, China

TAO DAI, Shenzhen University, Shenzhen, Guangdong, China

JIGANG BAO, Tsinghua University, Beijing, China

View all

Published: 08 May 2025

Citation in BibTeX format

WWW '25: The ACM Web Conference 2025
April 28 - May 2, 2025
Sydney NSW, Australia

Conference Sponsors:
SIGWEB

WWW '25: Companion Proceedings of the ACM on Web Conference 2025 (May 2025)
https://doi.org/10.1145/3701716.3715513
ISBN: 9798400713316

MERA: Mixture of Experts with Retrieval-Augmented Representation for Modeling Diversified Stock Patterns

YuJun Liu
Tsinghua University
Shenzhen, China
lyj697020@sina.com

Chen-Hui Song
E Fund Management Co., LTD.
Guangzhou, China
songchenhui@efunds.com.cn

Peiyuan Liu
Tsinghua University
Shenzhen, China
lpy23@mails.tsinghua.edu.cn

Naiqi Li*
Tsinghua University
Shenzhen, China
linaiqi@sz.tsinghua.edu.cn

Tao Dai*
ShenZhen University
Shenzhen, China
daitao.edu@gmail.com

Jigang Bao
Tsinghua University
Shenzhen, China
baojg19@mails.tsinghua.edu.cn

Yong Jiang
Tsinghua University
Shenzhen, China
jiangy@sz.tsinghua.edu.cn

Shu-Tao Xia
Tsinghua University
Shenzhen, China
xiast@sz.tsinghua.edu.cn

Abstract

Successful quantitative investment relies on accurate predictions of the future stock price. Deep learning-based solutions have recently demonstrated a superior ability to capture the intricate and nonlinear interactions among various market variables. However, most existing methods use the same parameters to fit all samples, without considering that the real stock market often exhibits multiple patterns. To alleviate this issue, we propose a novel module called Mixture of Experts with Retrieval-Augmented Representation (MERA). Essentially, MERA consists of a set of independent experts for differentiated modeling as well as a GateNet that dynamically allocates data of different patterns to the most suitable experts. The model backbone is responsible for learning the coarse-grained representations for all stock patterns. Then, each expert in the MERA module focuses on the specific pattern and performs a more fine-grained analysis. However, accurate data allocation remains challenging due to the lack of explicit pattern identifiers. To overcome this, MERA retrieves relevant samples using high-level representations from self-supervised pre-training. The label information of neighbor samples is promising discriminative signals to indicate the target stock pattern. Extensive experiments on real-world stock markets show significant improvements. Our code is released at https://github.com/chenchen1104/MERA.

CCS Concepts

• **Computing methodologies → Artificial intelligence**; • **Applied computing → Economics**.

Keywords

Stock Prediction, Mixture of Expert, Retrieval-Augmented Representation

ACM Reference Format:
YuJun Liu, Chen-Hui Song, Peiyuan Liu, Naiqi Li, Tao Dai, Jigang Bao, Yong Jiang, and Shu-Tao Xia. 2025. MERA: Mixture of Experts with Retrieval-Augmented Representation for Modeling Diversified Stock Patterns. In *Companion Proceedings of the ACM Web Conference 2025 (WWW Companion '25), April 28-May 2, 2025, Sydney, NSW, Australia.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3701716.3715513

1 Introduction

Stock market prediction is a crucial research area that empowers investors to make informed investment decisions and maximize investment profits. However, the dynamic and intricate nature of the stock market makes accurate forecasting challenging.

Over the past decade, deep learning techniques have emerged as a popular approach for stock prediction and quantitative investment. Unlike traditional statistical methods [3, 6], deep learning models [2, 8, 10, 11] excel at capturing complex and nonlinear relationships between various market variables. Despite the success of these methods, most of them involve supervised training on a single-branch model to fit all samples. This paradigm struggles to accommodate the diverse patterns inherent in real-world stock markets, resulting in suboptimal predictive performance. For example, during the early COVID-19 pandemic, high-performing stocks such as e-commerce and medical technology surged, driven by shifts in consumer behavior and market sentiment. In contrast, low-performing sectors like energy and aviation faced setbacks due to macroeconomic factors. This highlights the need for models capable of distinguishing and learning the diverse stock patterns.

This paper introduces a Sparse Mixture of Experts (SMoE) [5] as an extension module for capturing diverse stock patterns. SMoE consists of a GateNet and several independent experts. Each expert focuses on specific stock patterns, while GateNet assigns data with the particular pattern to the corresponding expert for processing. As

*Corresponding authors are Naiqi Li and Tao Dai.

WWW Companion '25, Sydney, NSW, Australia

ACM ISBN 979-8-4007-1331-6/2025/04
https://doi.org/10.1145/3701716.3715513

shown in Figure 1, unlike the vanilla single-branch model, our multi-branch expert model employs GateNet to dynamically allocate each stock data to a tailored subset of experts, enabling fine-grained analysis of stocks with varying performance.

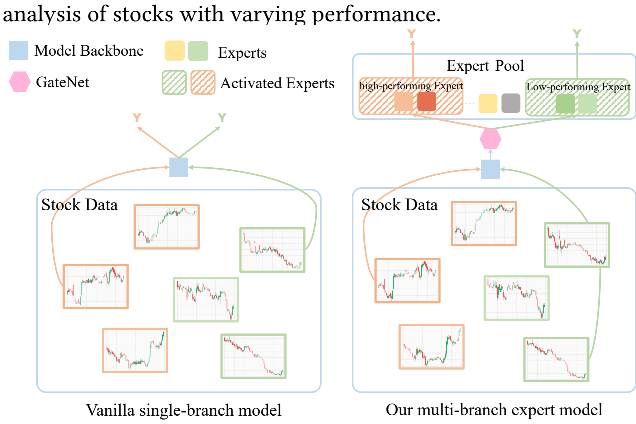

Figure 1: The comparison between the vanilla single-branch model and our multi-branch expert model.

However, accurate data allocation is still challenging due to the absence of explicit pattern identifiers. While stock patterns can be distinguished based on labels during the training phase, the lack of labels during testing complicates this process. The key challenge, therefore, is to develop a unified solution for distinguishing stock patterns during all phases. Recent retrieval-augmented (RA) techniques [1, 7] offer methods to estimate target stock patterns across all phases. The RA paradigm retrieves relevant information and uses it to enhance analysis for the target. When integrated into our framework, a self-supervised pre-training task is first used to transform raw noisy data into compact high-level representations. Then, for the target stock sample, similar samples are retrieved based on high-level representations, and an aggregation operation is performed over the retrieved samples. The aggregated label is a strong signal to indicate the target stock pattern, which is sent to GateNet for accurate data allocation. Meanwhile, both the aggregated feature and the target feature are directed to the specific expert for more fine-grained analysis.

In summary, our contributions are as follows: (1) Our proposed Mixture-of-Experts with Retrieval-Augmented representation module can capture multiple stock patterns, such as differentiated modeling for high-performing and low-performing stocks; (2) To enhance the accuracy of GateNet in distinguishing diverse stock patterns, we utilize self-supervised high-level representations to retrieve similar samples. Both the features and labels of the retrieved samples are strong signals to indicate the target stock pattern; (3) Extensive experiments conducted on the real-world stock market demonstrate the superiority of our approach over benchmarks and SOTAs. Furthermore, MERA is a plug-and-play module that can be introduced to any DL-based model.

2 Method

2.1 Problem Formulation

Formally, our objective is to develop a model $f_\theta(\cdot)$ that can predict future stock returns $\hat{y}_t^s$ from intra-day minute-level data x_t^s. Specifically, for each stock s on trading day t, there is a corresponding minute-level data $x_t^s = \{x_t^s[0], \cdots, x_t^s[k-1]\} \in \mathbf{R}^{c \times k}$, where $x_t^s[i]$ denotes the stock feature with dimension c for the i-th minute. There are k minutes of trading sessions per day. It includes information such as the opening price, closing price, highest price, lowest price, trading volume, turnover, and more. For the prediction label, we define it as the daily return ratio:

$$y_t^s = \frac{\text{close\_price}_{t+1}^s - \text{close\_price}_t^s}{\text{close\_price}_t^s}. \tag{1}$$

Since stock investment is to rank and select the most profitable stocks, we apply daily Z-score normalization to the return ratios to encode the labels with the rankings.

2.2 Self-supervised Pre-training

The stock market's raw data is often noisy, which challenges the subsequent modules' ability to retrieve similar samples effectively. To address this issue, we employ a pre-training strategy to transform the raw data into compact high-level representations.

Masked autoencoders are a widely used self-supervised pre-training method with many successful applications [4]. The core concept is straightforward: randomly remove portions of the input sequence and train the model to reconstruct the missing parts.

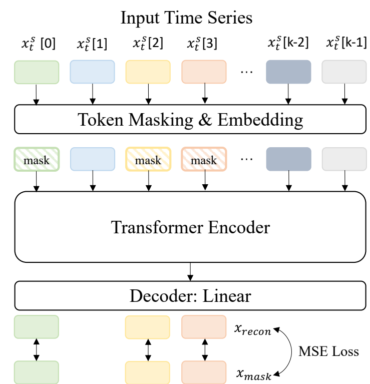

Figure 2: Masked self-supervised representation learning.

As shown in Figure 2, an autoencoder usually consists of an encoder that encodes inputs to high-level latent features and the decoder decodes latent features to reconstruct the input. Here, we choose a vanilla transformer as the model backbone. Given the input $x \in \mathbb{R}^{c*k}$ and a masking ratio m, we randomly select a subset of indices in each stock time series and mask the corresponding inputs with zeros. The model f is trained with MSE loss to make the reconstructed part x_{recon}, derived from the context of the visible part x_{vis}, as similar as possible to the original mask part x_{mask}. Given S as the size of the stock pool, it can be expressed as,

$$L = MSE(x_{recon}, x_{mask}) = \frac{1}{S}\sum_{i=1}^{S}(x_{recon_i} - x_{mask_i})^2. \tag{2}$$

2.3 Retrieval Augmentation

2.3.1 Overview. Figure 3 provides an overview of our proposed MERA module. We first construct the retrieval pool P based on pre-trained high-level representations of the training set $D = \{(x_1^1, y_1^1), \cdots, (x_T^S, y_T^S)\}$. Here, S represents the size of the stock pool, and T

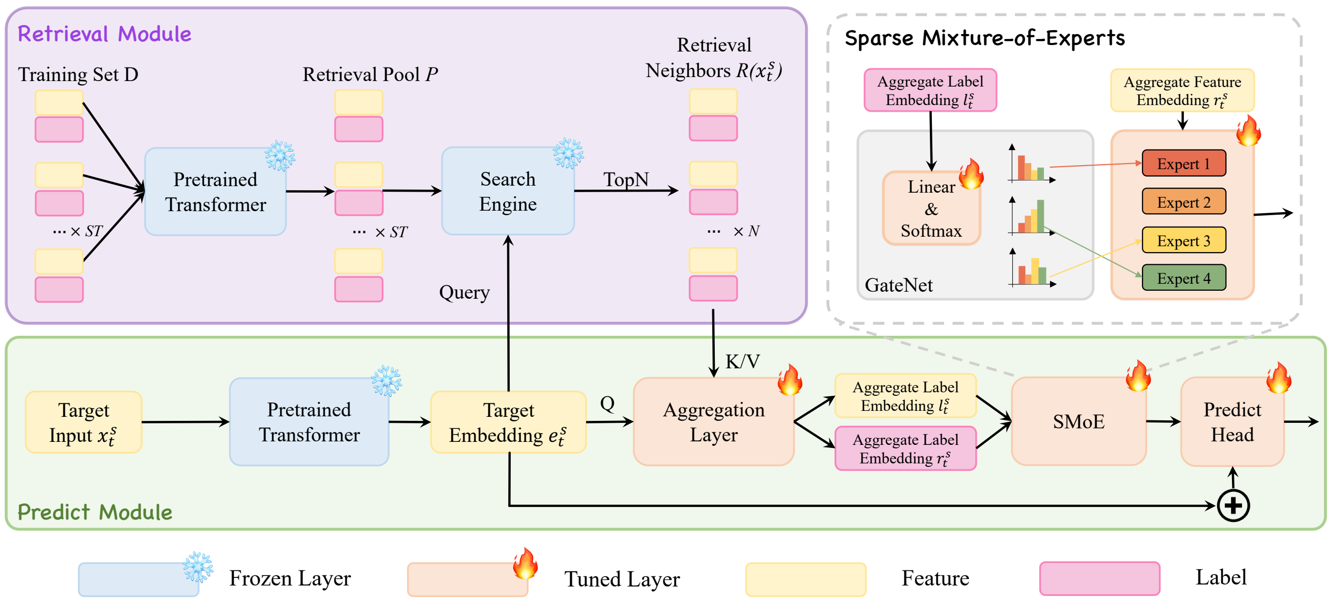

Figure 3: The overall architecture of our MERA Module.

includes the trading days in the training set. When a prediction request x_t^s is received, the pre-trained model backbone is first used to extract its feature embedding e_t^s, which will be regarded as a query to retrieve $TopN$ similar samples from the retrieval pool. Subsequently, the aggregation layer operates on features and labels, as both types of information are crucial.

The prediction module takes the representations of aggregated relevant samples and the target sample as the inputs. The aggregated label embedding l_t^s is sent to GateNet for accurate routing. The features (e_t^s, r_t^s) are combined and fed into the expert modules for more fine-grained analysis.

2.3.2 *Retrieval module.* The retrieval module is responsible for retrieving data samples that have relations with the target through nearest neighbor search.

When constructing the retrieval pool, we store both the pre-trained high-level representations and the label information. Notably, Continuous labels are discretized into categorical variables by dividing them into B ($B = 10$ as default) classes based on their numerical value. This is sufficient because label information is used to indicate stock patterns and determine which expert modules to activate, without requiring precise numerical values.

When a prediction request x_t^s is made, its feature embedding e_t^s is extracted as the query. The search engine retrieves $TopN$ most similar samples based on MSE metrics. These retrieved samples are then aggregated using a target-aware, order-invariant attention mechanism, which assigns a relevance-based scaling coefficient to each sample. The aggregated feature embedding r_t^s is formulated as:

$$\boldsymbol{r}_t^s = \sum_{i=1}^{N} \alpha_i \cdot \boldsymbol{s}_i^f, s_i \in \mathcal{R}\left(x_t^s\right). \tag{3}$$

where $\mathcal{R}\left(x_t^s\right)$ is the retrieval set of target sample x_t^s and s_i^f is the feature part of sample s_i. α_i is the attention weight of the i-th retrieved sample, which is defined as:

$$\alpha_i = \frac{\exp\left(\boldsymbol{x}_i^T \boldsymbol{W} \boldsymbol{x}_t^s\right)}{\sum_{j=1}^{N} \exp\left(\boldsymbol{x}_j^T \boldsymbol{W} \boldsymbol{x}_t^s\right)} \tag{4}$$

Where W is the attention layer parameter matrix. The label information is also aggregated using the same attention weights:

$$\boldsymbol{l}_t^s = \sum_{i=1}^{N} \alpha_i \cdot \boldsymbol{s}_i^l, s_i \in \mathcal{R}\left(x_t^s\right). \tag{5}$$

Finally, the target stock embedding e_t^s, the aggregated representations r_t^s, and l_t^s are fed into the prediction module to enhance the final predictions.

2.3.3 *Predict module.* The prediction module is responsible for calculating the final predictions based on the representations of the target sample and retrieved samples.

To capture the diversified stock patterns, the SMoE layer is first used for differentiated modeling. Given stock data, the output is a weighted sum of the outputs from k most relevant experts, with the weights determined by GateNet G. Only the k experts with the largest weights are activated, then unused experts need not be computed. As the key part, GateNet is responsible for selecting corresponding expert modules for data with specific stock patterns. Aggregated label embedding l_t^s is a strong signal to indicate the target pattern and is sent to GateNet for accurate allocation.

$$G(l_t^s) = TopK(Softmax(l_t^s \cdot W_g)). \tag{6}$$

The aggregated feature embedding r_t^s and target feature embedding e_t^s are fed into the activated expert modules for a more

Table 1: Quantitative comparisons among different methods in CSI300, CSI500 and CSI1000 dataset. Bold represent the optimal results.

Dataset	Method	Ranking Metrics				Portfolio Metrics		
		IC	ICIR	RankIC	RankIR	*Return*$^+$	*Return*$^-$	Return
CSI300	MLP	0.071	0.830	0.068	0.846	0.431	-0.772	0.602
	LightGBM	0.092	0.814	0.080	0.807	0.614	-0.911	0.763
	TCN	0.087	0.732	0.085	0.830	0.543	-0.974	0.759
	SFM	0.080	0.801	0.079	0.823	0.482	-0.922	0.702
	GRU	0.091	0.796	0.079	0.742	0.606	-0.909	0.757
	ALSTM	0.093	0.775	0.082	0.712	0.599	-0.988	0.794
	Transformer	0.095	0.821	0.094	0.848	0.658	-1.015	0.837
	Transformer+TRA	0.098	0.827	0.096	0.848	0.701	-1.047	0.874
	Transformer+CISP	0.102	0.832	0.100	0.850	0.707	-1.064	0.886
	Transformer+MERA	**0.107↑ 0.012**	**0.855↑ 0.034**	**0.106↑ 0.012**	**0.859↑ 0.011**	**0.726↑ 0.068**	**-1.104↑ 0.089**	**0.915 ↑ 0.078**
CSI500	MLP	0.080	0.962	0.078	1.011	0.701	-1.023	0.862
	LightGBM	0.100	0.997	0.093	0.178	0.874	-1.343	1.109
	SFM	0.089	0.888	0.092	1.203	0.799	-1.369	1.084
	TCN	0.100	0.969	0.096	1.184	0.847	-1.393	1.120
	GRU	0.101	0.970	0.090	1.024	0.798	-1.333	1.066
	ALSTM	0.104	0.981	0.101	1.142	0.868	-1.433	1.151
	Transformer	0.109	1.045	0.102	1.203	0.889	-1.524	1.207
	Transformer+TRA	0.114	1.089	0.106	1.250	1.012	-1.558	1.275
	Transformer+CISP	0.113	1.092	0.108	1.257	1.013	-1.556	1.280
	Transformer+MERA	**0.117↑ 0.008**	**1.097↑ 0.052**	**0.112↑ 0.010**	**1.262↑ 0.059**	**1.018↑ 0.129**	**-1.551↑ 0.027**	**1.285 ↑ 0.078**
CSI1000	MLP	0.093	1.211	0.086	1.241	1.171	-2.113	1.643
	LightGBM	0.120	1.365	0.093	1.152	1.498	-2.414	1.596
	SFM	0.113	1.396	0.092	1.232	1.315	-2.423	1.869
	TCN	0.121	1.393	0.099	1.200	1.521	-2.640	2.081
	GRU	0.119	1.411	0.097	1.132	1.504	-2.538	2.022
	ALSTM	0.122	1.371	0.090	1.017	1.479	-2.620	2.050
	Transformer	0.124	1.377	0.108	1.203	1.564	-2.601	2.083
	Transformer+TRA	0.125	1.432	0.108	1.190	1.662	-2.690	2.165
	Transformer+CISP	0.126	1.437	0.108	1.195	1.669	-2.685	2.170
	Transformer+MERA	**0.129↑ 0.005**	**1.444↑ 0.067**	**0.110↑ 0.002**	**1.256↑ 0.053**	**1.675↑ 0.111**	**-2.677↑ 0.076**	**2.176 ↑ 0.093**

fine-grained analysis. Here, we choose GRU as the expert architecture to construct an attention-GRU hybrid model. Model backbone employs self-attention layers for long-term dependencies. MERA module uses recurrent GRU layers for local processing. By simultaneously focusing on global information and local details, the model can perform a more comprehensive analysis.

Finally, with the residual connection, the stock embedding is encoded with both the information from the overall market and the personal information and is fed into a predictor for label regression. We use an MLP equipped with an activation function as the predictor, and the forecasting quality is measured by the MSE loss.

$$\hat{y_t^s} = MLP(\sum_{k=1}^{K} G_k(l_t^s) \cdot E_k((e_t^s, r_t^s))). \tag{7}$$

$$L_M = MSE(\hat{y}, y) = \frac{1}{S}\frac{1}{T}\sum_{s\in S}\sum_{t\in T} \|\hat{y}_t^s - y_t^s\|^2. \tag{8}$$

3 Experiment

3.1 Experiment Setups

To evaluate our approach, we conducted experiments using real-world intra-day minute-level data from three major Chinese A-share indexes: CSI300, CSI500, and CSI1000. These datasets are diverse and representative, offering a comprehensive assessment of model performance. We divided the dataset into a training set (2018/06/01 - 2022/05/31), a validation set (2022/06/01 - 2022/12/31), and a testing set (2023/01/01 - 2024/03/31) sets based on a time sequence.

Regarding the transformer architecture, we employ a standard two-layer transformer encoder with a feature dimension of 128. The MERA module is configured with 4 experts ($M = 4$), 1 of whom ($K = 1$) are activated for analysis. To ensure that relevant information can be obtained without introducing noise, we set the number of retrieved samples N to 10 as the default setting. We compared with existing methods (e.g. Transformer+TRA [9], Transformer+CISP [10]) using key evaluation metrics for predictive ability (IC, RankIC, ICIR, RankIR), and profitability (Long Return *Return*$^+$, Short Return *Return*$^-$, Long-Short Return *Return*).

3.2 Experimental Results

From the results presented in Table 1, it can be found that: (1) The MERA module significantly improves the performance of the backbone models, achieving SOTA results, thereby demonstrating the effectiveness of our proposed approach in enhancing stock prediction. (2) Portfolio metrics further reveal that the MERA module generates higher returns, which can be successfully applied in real-world investment scenarios to deliver superior returns.

3.3 Further Analysis

3.3.1 Visualization of data allocation. To demonstrate that our module adaptively allocates data of different patterns to the appropriate experts for analysis, figure 4 visualizes the logits of experts being selected for each label class. The x-axis represents expert ID, while the y-axis denotes label classes. The activation map exhibits extremely high sparsity. It can be observed that **label class 3/4/5** select **expert 3**, while the **label class 7/8/9** select **expert 0**, indicating that expert allocation is not random, and exhibits a strong correlation with stock patterns.

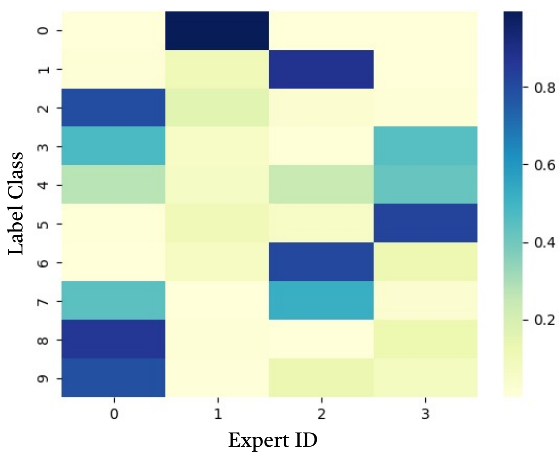

Figure 4: Visualization of the logits that experts are selected for each label class.

3.3.2 Ablation study. Our MERA module consists of two key components – SMoE and RA representations. We conducted ablation studies to verify the effectiveness of each component. The results show the following: (1) Comparing the performance of feeding x_t^s, r_t^s, and l_t^s directly into a single GRU layer without MoE architecture demonstrates that the multi-branch expert structure achieves differentiated modeling and improves prediction accuracy. (2) The model's performance drops when RA representations are omitted, highlighting their importance. (3) Using only the feature part of retrieved samples as input to GateNet significantly reduces performance without label information. Label embeddings serve as crucial discriminative features, aiding in identifying the stock pattern and allocating the data to the appropriate expert.

4 Conclusion

In this work, we propose a novel multi-branch module for capturing diverse stock patterns. Firstly, we design a self-supervised pretext task to pre-train the transformer. The learned high-level representations are stored in the retrieval pool, including not only the feature part but also the label part. Subsequently, for the target stock sample, we retrieve similar samples and perform an aggregation operation. The aggregated label part is sent to GateNet to indicate the target stock pattern and achieve accurate allocation adaptively, while the aggregated feature part is fed to activated experts for differentiated modeling. Extensive experiments demonstrate the superiority of our approach over benchmarks and SOTAs.

Table 2: Ablation study of SMoE and RA representations.

Model	SMoE	RA	label	IC	ICIR	Return
MERA	✓	✓	✓	0.107	0.855	0.915
		✓	✓	0.101	0.834	0.889
	✓	✓		0.102	0.828	0.871
	✓			0.102	0.823	0.863
				0.095	0.821	0.837

Acknowledgments

This work is supported in part by the National Natural Science Foundation of China, under Grant (62302309,62171248), Shenzhen Science and Technology Program (JCYJ20220818101014030, JCYJ20220818101012025), and the PCNL KEY project (PCL2023AS6-1).

References

[1] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. 2022. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*. PMLR, 2206–2240.
[2] Siyu Gao, Yunbo Wang, and Xiaokang Yang. 2023. Stockformer: Learning hybrid trading machines with predictive coding. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*. 4766–4774.
[3] Campbell R Harvey, Edward Hoyle, Russell Korgaonkar, Sandy Rattray, Matthew Sargaison, and Otto Van Hemert. 2018. The impact of volatility targeting. *Available at SSRN 3175538* (2018).
[4] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. 2022. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 16000–16009.
[5] Robert A Jacobs, Michael I Jordan, Steven J Nowlan, and Geoffrey E Hinton. 1991. Adaptive mixtures of local experts. *Neural computation* 3, 1 (1991), 79–87.
[6] Abby Y Kim, Yiuman Tse, and John K Wald. 2016. Time series momentum and volatility scaling. *Journal of Financial Markets* 30 (2016), 103–124.
[7] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems* 33 (2020), 9459–9474.
[8] Tong Li, Zhaoyang Liu, Yanyan Shen, Xue Wang, Haokun Chen, and Sen Huang. 2024. Master: Market-guided stock transformer for stock price forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 162–170.
[9] Hengxu Lin, Dong Zhou, Weiqing Liu, and Jiang Bian. 2021. Learning multiple stock trading patterns with temporal routing adaptor and optimal transport. In *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining*. 1017–1026.
[10] Chen-Hui Song, Xi Xiao, Bin Zhang, and Shu-Tao Xia. 2023. Follow the Will of the Market: A Context-Informed Drift-Aware Method for Stock Prediction. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*. 2311–2320.
[11] Shuo Sun, Xinrun Wang, Wanqi Xue, Xiaoxuan Lou, and Bo An. 2023. Mastering stock markets with efficient mixture of diversified trading experts. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2109–2119.